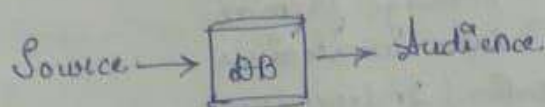


Ch-1

Databases → traditional db
 → Multimedia db
 → GIS: Geographical Information Systems
 → OLAP: Online Analytical processing
 → Realtime & active db

Database → has a source from which data is derived.
 → has some degree of interaction with real world
 → has an audience (or users) who actively interact

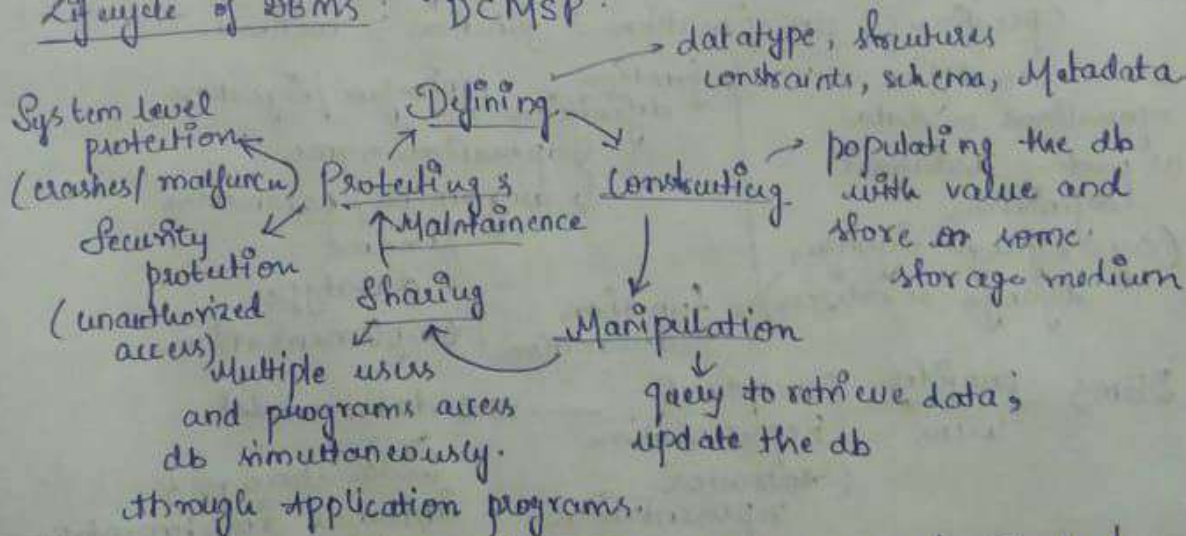


↑↓
 Interaction (Manipulation)

DBMS → general purpose software system

→ collection of programs that enables users to create & maintain a db

Lifecycle of DBMS: DCMSP



Name	USN	Class	Dept
Katrina	73	1	CS
Kaif	42	2	CD

← data elements / attributes / header

} data records / rows

Table or File

Characteristics of Database Approach

Why db approach > traditional file processing?

→ Self describing nature of a db system

- db not only stores data records, but also stores.
 - complete definition/description of its structure & constraints
- AKA Meta-Data
or DBMS Catalog
- structure of file/table
 - types/datatypers
 - storage format
 - data constraints

→ Program-Data Isolation/Independence
(Data abstraction)

• pgm-data independence: DBMS catalog/Metadata is stored separately from Access programs used to access the file.

• pgm-operation independence: What is an operation? An operation $\equiv$ function $\equiv$ method.

users can define operations on data as part of database definition

(e.g. Django signals, Django serializers)

function definition / Interface / Signature

- ↳ operation name
- ↳ arguments / parameters
 - ↳ name
 - ↳ datatype

function declaration / Implementation

DBMS provides users

Conceptual Representation
(Abstracted representation)

→ AKA data model

hides storage & internal implementation details

→ Supports Multiple Views of Data

- A db typically has multiple users.
- Each user might be interested only in a subset of the db or virtual data (meaning: derived data from the db which is not explicitly stored, but is only used for presentation).

- Sharing of Data and Multiuser Transaction processing
- multiple users access data at same time.
- Concurrent control / update of db (eg: Train Reservation).
- OLAP and OLTP.

What is transaction? txn

Transaction $\rightarrow$ executing program / process
 $\rightarrow$ Includes 1 or more db access (such as read or update, etc)

- A db must use foll. properties:

* Isolation: ensure each txn appears to execute in isolation though lots of txn may be executing concurrently

* Atomicity: All db op in a txn are executed or none are.

Eg: Payment gateway.

Actors on the Scene

$\rightarrow$ refers to people involved in the design, use and maintenance of a large database

1) Database Administrator (DBA)

Responsibilities:

- Authorizing access to the db
- Coordinate & monitor use of the db
- Acquire hardware & software resources
- Security breaches & poor system response time.

2) Database Designer

- Identify data to be stored.
- Choose data structures to represent & store data
- Develop different views of the db for different groups of users.

3) End users

- require access to the db for querying, updating & generating reports

* Types of end users :

- * Casual end users : Sophisticated db query language
- * Naive / parametric : constantly query & update
- * Sophisticated : Engineers, Scientists, business analysts
- * Standalone : maintain personal db with menu or gui based apps

4) Software Engineers

- System Analysts : determine requirement of end users
- Application programmers : Implement the specifications from system analysts as programs - test, debug, document & maintain

Workers behind the Scene

1) DBMS System designers & Implementers :

- design & implement DBMS modules & interfaces as a software package
- * Modules for catalog implementation
- * Query Language processing * Control concurrency
- * Interface processing * Access & Buffer data
- * Data recovery & security
- * Interface with OS, Compilers & other pgm languages

2) Tool developers :

- design and implement tools
- * database design * performance monitoring
- * natural language / graphical interfaces
- * prototyping, simulation, test data generation

- 3) Operator & Maintenance personnel.
• running & maintenance of hardware & software environment of the db.

Advantages of using DBMS approach

- Controlling Redundancy.
 - duplication of effort
 - wastage of storage
 - Inconsistent
 - data normalization
- Restricting unauthorized access.
 - provide security & authorization subsystem.
- persistent storage of program objects.
 - Impedance mismatch problem : data structures provided by the DBMS were incompatible with another pgm language structures.
- provide storage structure & search techniques for efficient query processing.
 - use Indexes for disk search (tree data structure or has data)
 - provide buffering/caching
- provide Backup & Recovery
- provide multiple user interfaces depending on on scene or off stage actors.
- Represent complex Relationships among data.
- Enforce Integrity Constraints (Later).
- permitting inferencing & actions using rules.
 - deductive dbms
 - rules
 - triggers
 - stored procedures